

 (Fluid Flow Through Fluidized Beds)

Pre-Lab Plan
Unit Operations Lab 1 
Date: 9/5/19

Group 4, Thursday, PM section

Team Member Names:

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1. Experimental Objective

The objective of this lab is to study the behavior of a bed of two types of small particles (one being fine silica sand and the other being ion exchange resin beads). We will also measure the gas pressure drop across the bed of particles as a function of the gas velocity.

2. Experimental

2.1 Apparatus

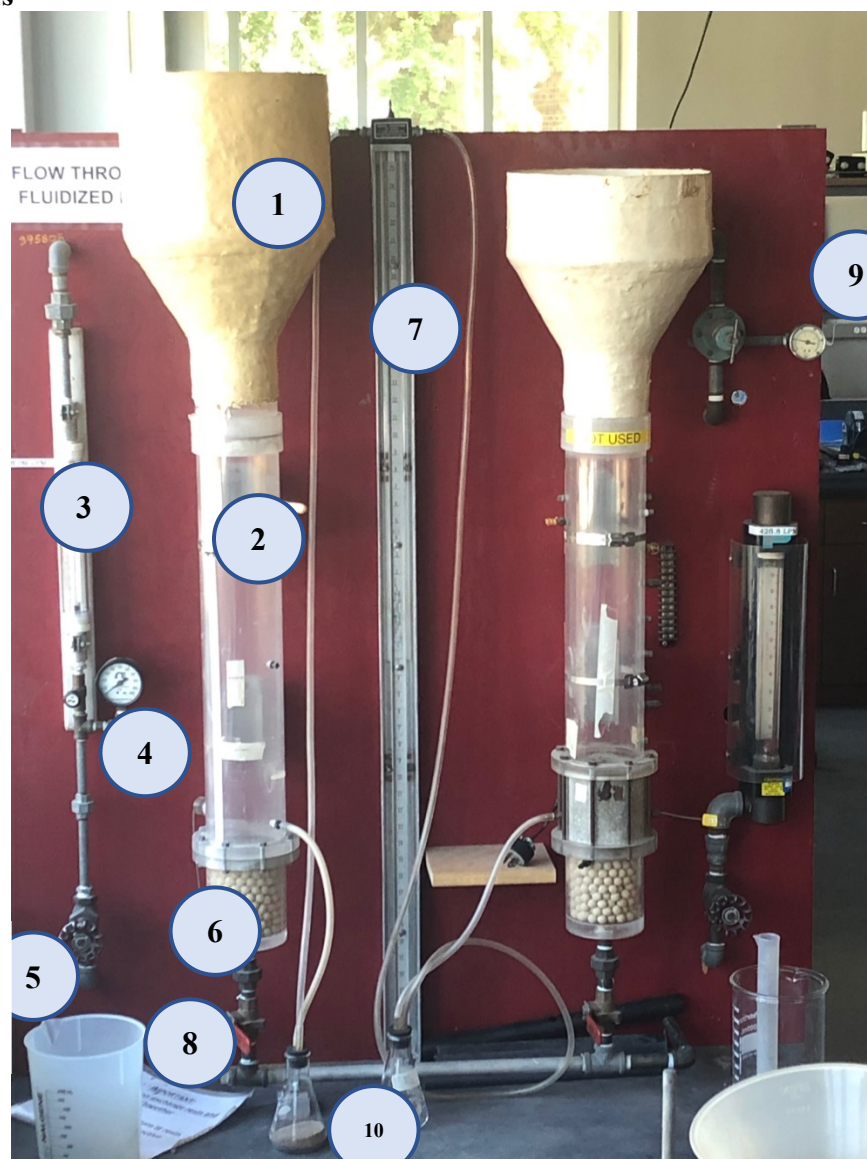


Image 1: This picture illustrates the fluidized bed experiment and apparatus. For a detailed description of each apparatus, refer to table 1.

Table #1:

Number	Component Name	Description
1	Disengaging section	The section of the fluidized bed that allows for the exit of the air
2	Column	Holds the particle material, area where the bed is fluidized
3	Flow meter	Measures the volumetric flow rate of air
4	Pressure gauge	Measures the pressure of air passing to the column
5	Flow adjustment valve	Controls the flow of air
6	Air distributor beads and plate	Distributes the inlet air across the surface area of the column
7	U-Type Manometer	Measures pressure acting on a column of fluid
8	Ball valve	Controls air to the column
9	Inlet pressure gauge	Measures pressure of the air to the process
10	Trap	Catches sand/resin before entering the manometer

This table is a compilation of the components from image 1 along with the descriptions of said component and how they function within the overall process.

2.2 Materials and Supplies

Table #2:

Index	Material/Supply Name	Description
1	Silica sand	Material 1 used for the fluidized bed
2	Ion exchange resin beads	Material 2 used for the fluidized bed
3	Air	Used to create the fluidized bed
4	Sieve tray	Tray used to measure particle size
5	Graduated cylinder	A volume measuring device

This table is a compilation of the materials and supplies along with the descriptions of said material/supply and how they function within the overall process.

2.3 Experimental Plan

Fluidized Bed Procedures

1. Familiarize with lab setup using the lab manual provided
2. Obtain the silica sand and ion exchange resin beads used for creating the fluidized bed
3. Pack the bed with 6 inches of silica sand
4. Slowly open the air valve and record the pressure drop as a function of gas flow rate
5. Continue to increase flow until the minimum fluidization point is surpassed
6. Decrease the gas velocity and record the pressure drop as velocity of gas is decreased to zero
7. Take measurements at static heights of 6 and 10 inches with the bed material
8. Plot the measured pressure drop per unit bead height as a function of the gas velocity for both increasing and decreasing flowrates
9. Repeat steps 3 – 8 using ion exchange resin beads

Particle Size Determination Procedure

1. Acquire sieves from cabinet under experiment (Diameter range: 0.001 – 0.0787 in.)
2. Measure the mass of a known amount of sand and place in top sieve
3. Shake sieves to separate out sand particles by size
4. Measure and record the mass of sand left behind in each sieve
5. Using the mass of sand from each sieve determine the distribution of each particle size and the average particle size
6. Repeat steps 2 – 5 for the ion exchange resin beads

Void Fraction Determination Procedure

1. Fill graduated cylinder with 50 ml of sand
2. Fill graduated cylinder with water until the water fills to the 50 ml line
3. Record the volume of water added (this is the void volume)
4. Use the void volume to calculate the void fraction
5. Repeat 1 – 4 with ion exchange resin beads

The independent variable in this experiment is the air flow rate supplied to the column. The dependent variable is pressure drop.

For this experiment there is no statistical or calibration methods needed. Uncertainty can be minimized by increasing the number of sieves used to ensure the particle size is accurately measured.

Data from this experiment will be analyzed in excel. The goodness of fit from the data will be determined using excel.

Using the Ergun equation (Equation 1), the pressure drop can be accurately calculated by measuring the various experimental parameters. It is also known that for small particles and for laminar flow (Reynolds number less than 1) gives a simplified equation (Equation 2) for calculating the minimum fluidization velocity. It is expected that the measured minimum fluidization velocity is close to that of the theoretical one.

$$\frac{\Delta P}{L} = 150 \bar{V}_0 \frac{\mu}{\Phi_s^2 D_p^2} \left(\frac{(1-\epsilon)^2}{\epsilon^3} \right) + \frac{1.75 \rho \bar{V}_0^2}{\Phi_s D_p} \left(\frac{1-\epsilon}{\epsilon^3} \right) \quad \text{Equation 1}$$

$$\bar{V}_{0M} \approx \frac{g(\rho_p - \rho)}{150\mu} \frac{\epsilon_M^3}{(1-\epsilon_M)} \Phi_s^2 D_p^2 \quad \text{Equation 2}$$

Project Safety Evaluation

Project Title: Fluid Flow Through Fluidized Beds

Date: 9/5/19

	<i>Yes/No</i>	
Potential electrical hazards?	No	If yes, identify conditions:
Potential mechanical hazards?	No	If yes, identify conditions:
Potential pressure hazards?	Yes	If yes, identify conditions: Columns and lines are under pressure. Potential for pressure breach
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	No	If yes, specify type:
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: None for this experiment
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	ambient	none
Pressure (psia)	40 psi	Inlet pressure, pressure to flowmeter, pressure to column
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	no	

I have read relevant background material for the Unit Operations Laboratory entitled: “Fluid Flow Through Fluidized Beds” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Jake Proctor</u> signature of team member	<u>9/5/2019</u> date
<u>Courteney Fabbri</u> signature of team member	<u>9/5/2019</u> date
<u>Milos Markovic</u> signature of team member	<u>9/5/2019</u> date
<u>Erica Caruso</u> signature of team member	<u>9/5/2019</u> date